MUHAMMAD MUTAHHAR BAIG

ELECTRONICS ENGINEER

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ABOUT ME

Electronics Engineer from UET Taxila, specialized in Simulations, AI, ML, and DL. Proficient in MATLAB signal processing, Python-based deep learning (TensorFlow, MediaPipe etc), and embedded platforms (ESP, Arduino). Winner of the 2025 Electronics Department Open House competition. Currently I have 1 year of professional hands-on experience in my field.

WORK EXPERIENCE

MachaDev Engineering, Air university Islamabad **July,2025-November,2025**

- Designing and development of embedded systems and electronic hardware component.
- Testing and maintaining firmware for microcontrollers and microprocessors including CH32, Arduino, ESP32, PI zero, PI 5, RK3566, RK3576, RK3588, STM32 etc.
- Developed real-time device software using C/C++, Python, MATLAB environments.
- Training, optimizing, and deploying machine learning and deep-learning models to edge devices and microprocessors.
- Performing model compression and optimization (quantization, pruning, TensorFlow Lite, ONNX, RKNN) for constrained hardware.
- Implementing and image-processing pipelines for object detection, gesture recognition, and sign-language.

Khayal Incotech. **December,2025-Present**

- Deployed and optimized Speech-to-Text pipelines (Whisper, Faster-Whisper, Deepgram) on Hugging Face Spaces, achieving 33-minute processing time for 1-hour audio with seamless REST API integration for production-ready scalability.
- Engineered an ESP32-S3 audio acquisition system featuring I2S MEMS microphones, SD card storage, and dual wireless transfer protocols (BLE & Wi-Fi Direct), enabling smooth mobile app integration.
- Conducted a comprehensive benchmark study across 8+ STT APIs (AssemblyAI, Groq, OpenAI, Google Cloud), evaluating speed, accuracy, and cost metrics to deliver data-driven production deployment recommendations.
- Developed embedded firmware for an interactive children's audio device on ESP32-S3, implementing a multi-tasking architecture with FreeRTOS (3 concurrent tasks), interrupt-driven rotary encoder navigation, and real-time I2S audio streaming.
- Architected a hierarchical navigation and audio management system with SD card file handling, supporting interruptible playback, context-aware voice announcements, and hardware-level real-time volume control via I2S peripheral.
- R&D for a pocket-sized AI device, exploring miniaturized hardware and software architectures to deliver accessible, on-device AI experiences in a compact form factor.

INTERNSHIPS

PAC Kamra Internship | Internship **Aug,2024- Aug,2024**

- Gained hands-on exposure to data preprocessing techniques and analysis methodologies used in predictive maintenance systems through observation and guidance from senior engineers.
- Observed and documented performance evaluation processes for AI-based fault detection models, understanding key metrics such as accuracy, precision, and recall.
- Developed foundational understanding of predictive maintenance workflows and the practical application of machine learning in industrial settings at PAC Kamra.

NICAT (National Incubation Center for Agri-Tech) NASTP, Pakistan | Internship **Aug,2024-Feb,2025**

- Developed and optimized an AI-powered weed detection system for precision agriculture, improving crop management efficiency and reducing herbicide usage through targeted application.
- Implemented model optimization techniques to enable real-time deployment on edge devices, ensuring system viability for field conditions with limited computational resources.
- Developed initial stage AI-powered weed detection system to enhance precision agriculture.
- My final year project has linked with this industry.

TECHNICAL SKILLS

- **Programming Languages:** Python, C/C++, MATLAB, Verilog
- **AI:** ML, DL, LLMs, Prompt Engineering, SQL, Gen AI, PyTorch, TensorFlow, TensorFlow Lite, OpenCV, MediaPipe, NumPy, Pandas, Model Training, Model Deployment, Feature Extraction, Model Optimization, Quantization, Pruning, ONNX, RKNN.
- **Embedded Systems:** Arduino, ESP32, Raspberry Pi (Zero, 5), FPGA, Pixhawk, STM32, CH32, Rockchips (RK3566, RK3576, RK3588), J-Link
- **Development Tools**: ESP-IDF, MounRiver Studio, Arduino IDE, MATLAB, NRF connect
- **Signal Processing:** ECG, PCG, Medical Imaging, Digital Signal Processing, Image Processing
- **Hardware:** Microcontroller Programming, Microprocessor Programming, Circuit Design, Sensor Integration

EDUCATION

Bachelor's Electronics Engineering | UET, Taxila November, 2021- June, 2025

FINAL YEAR PROJECT

Design & Development of Agri-Rover for Crop Management (FYP)

- Conceptualized and designed the mechanical and electronic architecture of an AI-powered Agri-rover,
- selecting Sunroof NSA-R2 Motors, BTS7960 drivers, and a 6S LiPo battery to balance torque, speed, and
- runtime for up to 10 kg payloads.
- Developed real-time crop monitoring algorithms using OpenCV/TensorFlow, enabling automated
- detection of onion, garlic, and potato plants with >90 % accuracy under variable lighting and occlusion
- conditions.

- Integrated multi-sensor fusion, combining RGB camera data, soil moisture probes, and NPK nutrient sensors, to assess plant health and soil fertility.
- model centered on sustainable practices and pollution reduction.
- Completely Autonomous, GPS navigation, used three modes (Auto, manual & guided) for easy access.

CERTIFICATIONS

- Hafiz e Quran
- IEEE Student Member (Consumer Technology Society, Engineering in Medicine and Biology Society)
- Winner, Open House 2025, Electronics Engineering Department
- Final Year Project selected for All Punjab Expo Competition 2025
- Submitted 1 conference paper.